

**Emergence of Bid-Ask Spread Dynamics in a Limit Order
Book Simulator : Effects of Liquidity, Order-Flow
Imbalance and Cancellation Intensity**

Varad Pote

Undergraduate Student, Computer Science

Golden Gate University

Abstract

In this study, there has been the use of a stochastic agent-based simulator of a continuous double-auction limit order book (LOB), with a dynamically evolving mid-price. Limit orders in this simulator follow an exponential offset from the reference price. Meanwhile, all the market orders target visible depth, and cancellations are programmed to focus on stale orders that are far from mid prices. To analyze what actually affects the bid-ask spread, the simulator was run with 100,000 initial orders and 20,000 steps along with three parameter sweeps which further analyzes the effect of liquidity, order-flow imbalance and cancellations. The major finding from the simulation was about the strong negative connection between spread and liquidity ($r \approx -0.33$). Specifically, when the liquidity is low the spread tends to jump significantly high from 0.1914 to 0.4769 in balanced market conditions. The simulation also showcases that order-flow imbalance and higher cancellation rates further make the spread much more volatile. The results obtained from this simulation experiment match the real-world stylized facts, therefore proving the model to be reliable and transparent. Overall, the framework which is used for the experimentation purpose acts as a controlled environment like a lab to study market microstructure and liquidity dynamics.

Keywords: limit order book, bid-ask spread, liquidity, order imbalance, market microstructure, agent-based simulation

Introduction

In the modern electronic markets, the bid-ask spread is the most basic parameter to measure liquidity and transaction costs (Madhavan, 2000). Because of that even a tiny shift in the spread ripples through the market by impacting everything from price discovery to overall efficiency. For years, economists and market microstructure researchers have focused on finding what actually causes spread to form and how it reacts to liquidity, order-flow imbalance and cancellations (Amihud et al., 2005; Chordia et al., 2002).

Most studies from the past have consistently shown how market conditions such as liquidity, order-flow imbalance and cancellations are closely linked with spread. Liquidity is the factor that usually dictates how wide the spread should be (Amihud et al., 2005), while order-flow imbalance puts temporary but significant pressure on all these numbers (Chordia et al., 2002). High-frequency cancellation rates, which is now one of the most dominant features of electronic markets further complicates the situation (Gould et al., 2013). However the issue that we face using real world data is that it is very messy making it difficult in proving what is causing what. There are so many things that are simultaneously occurring in the market which makes it nearly impossible to isolate just one variable.

To overcome all these hurdles with real world market data researchers came up with these agent based simulations to create controlled environments for experimentation.

Zero-intelligence models have often shown that just purely random trading can replicate the real price behaviour quite well (Gode & Sunder, 1993; Smith et al., 2003). All the recent models of stochastic limit order books which were often framed within the language of queuing theory have successfully replicated stylized facts that were observed in the real markets such as heavy-tailed spread distributions and long memory in order flows (Cont et al., 2010; Bouchaud et al., 2002; Cont & Müller, 2021).

However, many of the models have big flaws : they either keep the mid price static or lack systematic parameter sweeps or use rules which are too simple to replicate the real world market. The simulator used in this particular study has several realistic features making it closer to the real world market. The features are : Firstly, a dynamically changing mid-price driven by a Gaussian random walk, then an exponential offset distributions for limit-order placement that generate natural depth profiles, further it has market orders that are explicitly sized to consume visible depth, and lastly it has got stale-biased cancellations that preferentially remove orders far from the current mid-price.

The primary objective of this research is to study how liquidity, order-flow imbalance and cancellation intensity affects different dynamics of bid-ask spread. With the help of five major experiments and three parameter sweeps, the study produces results in a fully reproducible computational environment. In doing so, it provides a transparent lab for testing market-microstructure hypotheses and bridges the gap between empirical observations and theoretical models.

The remainder of the paper is organized as follows. The next section reviews the relevant literature on limit order book dynamics and spread formation. This is followed by a detailed description of the simulator architecture and experimental design. Results are then presented, highlighting key statistical findings and visualizations. The paper concludes with a discussion of implications, limitations, and directions for future research.

Literature Review

The study of limit order books and bid-ask spread has grown over the years, it has entered into several interconnected streams such as economics, market microstructure and the physics

behind the complex systems. From the very early days the major focus has remained on how liquidity affects the price of an asset and the cost to transact that. Amihud et al. (2005) provided us with the fact that liquidity is a risk factor, with bid-ask spread serving as the primary way to measure liquidity. Some empirical research confirmed that the spread tends to widen with low liquidity and high order-flow imbalance (Chordia et al., 2002; Madhavan, 2000).

At the same time researchers began studying deeply the actual micro structure of electronic order books. By analyzing high frequency data Bouchaud et al (2002) found several stylized facts such as long-range correlations in order-flow, heavy tailed distribution of order sizes and the tendency of the mean to exhibit persistent but mean reverting behaviour. See all these patterns in the real markets is what pushed the need for more realistic models of electronic order books.

One of the major breakthroughs was the use of zero intelligence models to see if simple rules were able to explain the market rather than complex strategies . Gode and Sunder (1993) showed that even when the traders act randomly, the market can still be efficient and find the right place. Smith et al. (2003) took this work further and showcased that the way prices and spreads are distributed happen naturally when the random orders interact with the book.

Building on the ideas discussed earlier, some models which came out started treating the LOB like a queueing system or a group of particles. Cont et al. (2010) built a model where limit orders arrive randomly which is called Poisson's process and are placed at various distances from their best quotes. Their model was successful in mimicking the real-world data, especially the depth of the order book and how prices change. Cont (2001) also highlighted the importance of stylized facts in simulators, which afterwards became the benchmarks for all simulations to check whether it's realistic or not.

More recently the researchers have added more layers to make it more realistic. For example, Cont and Müller (2021) developed a stochastic differential equations framework to create a link between individual orders and overall price moves. Gould et al. (2013) also put together an extensive review over many surveys that bring together different theories, data and simulation methods. Huang et al. (2013) and other studies have also conducted studies which were majorly focused on the cancellation activities, its main focus was to see how cancellations affect spread volatility and liquidity.

Even with all the progress made, there are still major gaps in the research. Many of the earlier existing models have either fixed the mid-price, or over simplify the order placement distribution or fail to test how cancellations and liquidity actually causes the spread to change. On top of this, only a few researchers allow their code to be open source to replicate it or build results on to that easily.

This study tries to solve all the above problems by building a transparent, stochastic LOB simulator that incorporates a dynamically changing mid-price with the help of random walk, exponential offset distributions for realistic depth profiles, market orders which are sized to consume visible depth, and stale-biased cancellation rules. This simulation approach has created a lab to finally see how liquidity, order-flow imbalance and cancellations interact to the creation of spread. This is all done by running five major experiments and three parameter sweeps.

Methodology

3.1 Simulator Architecture

This simulation was completely built using python, also it was designed fully as a stochastic, discrete-time model of a limit order book (LOB). The heart of the code is an OrderBook class; then SortedList from sortedcontainers library to keep all the price levels organized, which is then combined with dictionaries to ensure order tracking remains fast($O(1)$ complexity). Then it has got a MatchingEngine which is based on standard price-time priority, along with a main Simulator class that does all the heavy lifting in this experiment-from generating and matching orders to managing cancellations and saving the data. Each and every part of this architecture is clear and modular so that all the experiments carried out are easy to replicate.

3.2 Dynamic Mid-Price Evolution

The reference (mid) price follows a Gaussian random walk:

$$P_t = P_{t-1} + \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, \sigma_{\text{mid}}),$$

With

$$\sigma_{\text{mid}} = 0.02$$

After each trade, a small momentum term is added:

$$P_t \leftarrow 0.95 \cdot P_t + 0.05 \cdot p_{\text{last trade}}$$

This introduces realistic trade-impact feedback while keeping the process stationary

3.3 Order Placement Rules

To place limit orders, the simulation uses an exponential offset relative to the current mid price P_t :

$$d \sim \text{Exp}(\sigma),$$

where $\sigma_{\text{order}} = 0.30$ during simulation and $\sigma_{\text{initial}} = 2.0$ for initial book seeding.

Buy orders are placed at $P_t - d$ and sell orders at $P_t + d$, then rounded all these values to a 0.01 tick size to match standard market increments. This method helps us to mimic the real world scenarios where the liquidity is thickest near the mid-price and reduces as it goes far off, which is a well known stylized fact (Bouchaud et al., 2002; Cont et al., 2010).

On the other hand market orders occur at a 50% probability at each step. This simulation has a size range of 10-50 units per order. The sizes are intentionally kept large enough to consume visible depth at the best bid or ask and then widen the spread.

3.4 Matching Engine and Order Book Mechanics

The matching engine that is being used in this simulation follows a set price time priority rule which is used in most major exchanges. When a limit order comes in, it checks whether it can match with prices on the other side and if not it lies in the book. All these processes take place instantly. The code also contains an OrderBook class to keep all the bids and ask values separately. This helps in for very fast lookups, specifically $O(\log n)$ for finding the best prices, while the total liquidity and all the order counts are tracked constantly in $O(1)$ time complexity.

3.5 Cancellation Mechanism

The code had an adjustable cancellation rate but it was still using the default of 5 cancellations per simulation step for all the main runs. The mechanism preferentially targets “stale” orders using the probability:

$$p_{\text{cancel}} = \min \left(1, 0.1 + \frac{|p_{\text{order}} - P_t|}{\text{stale_distance}} \right),$$

with `stale_distance = 1.0`. The idea here is simple, in real markets when orders are sitting far away from the current price trends, it’s very unlikely to be fulfilled so the traders more than often pull those backs to try those somewhere else.

3.6 Simulation Procedure

The main simulation loop proceeds as follows for each step $t=1$ to 20,000:

```

for t in 1 to num_steps:
    # 1. Evolve mid-price
    ref_price += Normal(0, mid_price_volatility)

    # 2. Place new limit orders
    for i in 1 to limit_orders_per_step:
        order = generate_limit_order(ref_price)
        process_order(order)          # matching may occur

    # 3. Possibly place market order
    if random() < market_order_prob:
        market_order = generate_market_order()
        process_order(market_order)

    # 4. Perform stale-biased cancellations
    cancel_stale_orders()

    # 5. Record full LOB state
    record_state(t)

```

3.7 Data Collection and Statistical Analysis

During the complete simulation process, it was made sure that it records the full state of the order book at every single step including key metrics such as best bid/ask, the spread and the total volume. Once all the experiments were done, Pandas library(version 2.x) was used to process the data and later Matplotlib was used for the visualization purpose. Before deriving any summary statistics the data was thoroughly cleaned without any missing or NA spread values to increase efficacy . Pearson correlations tests were then run to see the relationship of spread with both liquidity and order-flow imbalance. In the interest of full reproducibility, I have made all the raw data, Python scripts and parameter settings available upon request.

Results

Across the five main experiments and three parameters sweep, the simulation generated a substantial dataset of over 20000 observations per run. The final analysis was only conducted on valid spread values, making sure to filter out any of the NA results that occurred during the initial formation of the book. The summary of all these findings can be found in Table 1.

While analyzing the summary we can see the spread was tightest in the buyer-heavy and seller-heavy scenarios coming in at 0.1329 and 0.1411 respectively. In contrast to that we saw the spread increase significantly to 0.4769- which is more than twice as wide as the baseline balanced market. The spread was not only wider in the low liquidity settings but it was also very volatile seen by high standard deviation. It was also observed that the spread increases as soon as the cancellations increase up to 0.2362 which is clearly higher than the balanced market conditions. It was seen that the trading activity stopped during the low liquidity setting as seen by lack of depth in the order book.

Figure 1 displays how the bid-ask spread has evolved over the five experiments. Once the initial seeding was passed the balanced markets and the high cancellation markets stayed fairly steady fluctuating between 0.15 and 0.25. However the low liquidity phase was much more erratic and showed frequent and aggressive spikes. While looking at the distribution (Figure 2) every experiment showed a rightly skewed pattern, especially low liquidity market had a much heavier tail, which meant wider spreads were quite common.

The parameter sweeps provided additional insight into the sensitivity of the spread. As we can see in figure 3 the mean spread dropped when the initial liquidity was increased. It dropped from about 0.8 at 500 orders down to just 0.19 at 100,000 orders which confirms the strong inverse between the cost and liquidity. Figure 4 showcases that when the buy ratio hits extremes like 0.1 or 0.9 the spread gets wider and more volatile, however the impact was not as dominant as liquidity. In the end figure 5 shows a direct positive link between the cancellations and the spread. As we scaled the cancellations from 0 to 30 per step, the steady widening of spread was found.

As we can see in figure 6 the mid-price paths move in a natural random walk pattern, though there are sharp jumps occasionally when a large order hits the books. In the balanced and high cancellation setups, it was clearly seen the price stayed fairly anchored around the starting point of 100 , however in several runs which are either dominated by buyers or sellers the price begins to steadily drift in that particular direction.

Ultimately, all these results proved that liquidity is the biggest driver of the bid-ask spread whereas cancellation rates and order-flow imbalance played smaller roles. It was also notably the simulation was able to mimic the real market with the rightly skewed spread data and the clear negative link between the spread and liquidity.

Discussion

The simulator that was built for this study did exactly what it was supposed to do. It mimicked the real markets and the spread behaviour to give a clear outlook on how liquidity, order-flow imbalance and cancellations actually drive the markets. Just as it was expected from all the literature around the spread, liquidity turned out to be the single most important factor to determine the width of spread. Specifically when the liquidity was lowered, it was observed that the mean spread was more than doubled jumping from 0.1714 to 0.4769, while the overall market became more volatile during the same time. The findings were closely similar to Amihud et al. (2005) who argued liquidity being the key risk factor and Gould et al. (2013) who pointed to market depth as the main driver of trading costs. The consistent negative correlation which was found ($r \approx -0.33$) really proves the point; in a controlled environment when the liquidity increases transaction costs go down.

The experiments also showcased that order-flow imbalance too played a role in the behaviour of spread, though it was not very dominant as liquidity. Interestingly, the average spread in both the buyer heavy as well as the seller heavy runs were a bit narrower than the balanced one, but they were far more volatile and prone to extreme jumps. This lines up exactly with what Chordia et al. (2002) saw in the real world market data where a heavy tilt in order flow creates temporary spikes in the spread. The parameter sweeps which were conducted also confirmed this, when the buy ratios hit extremes like 0.1 or 0.9 both, the average spread as well as the standard deviation climbed, suggesting that one-sided order flow reduces the depth on one side of the book and increases spread fluctuations.

It was very clear from the finding that the spread widens with higher cancellation. In the high cancellation runs the mean spread grew to 0.2362, which is a very noticeable jump from 0.1914 which was seen in baseline balance, also the cancellation-rate sweep confirmed a

monotonic increase in spread as the number of cancellations per step increased. This exactly supports the idea which was put forward by Huang et al. (2013), that aggressive cancellations thin the market depth which then in turn drives up the trading costs. The stale-biased cancellation rule which was implemented here appears to have captured an important realistic mechanism: traders preferentially remove orders that become mispriced relative to the moving mid-price, thereby contributing to spread persistence.

The simulator actually managed to replicate some of the stylized facts seen in the real world markets. For instance the skewed spread distribution in Figure 2 is a direct match for the empirical patterns that were shown by Bouchad et al. (2002) and Cont et al. (2010) . All the price paths shown were also realistic: they followed a standard random walk but had those sharp sudden jumps whenever a big order cleared the book just like the models described by Cont and Müller (2021). All these similarities really validate the choices that were made to perform this study especially using an exponential distribution for order placement and dynamically changing mid-price—ultimately this showcases that even a single zero intelligence setup can produce the complex real world behaviour such in financial systems (Gode & Sunder, 1993; Smith et al., 2003).

Thus all these results prove why a controlled environment is necessary and useful for studying market microstructure. By using controlled lab environments for this study, to isolate one variable at a time was valuable to cut all other confounding factors that makes the real world data so difficult to analyze (Madhavan, 2000). The open source nature of this data allows other researchers to replicate the results or even calibrate it against different sets of real world LOB data.

Limitations

There are few limitations to this model that should be noted. First, the model employs zero-intelligence agents; so all the strategic behaviours, learning or information based trading is not present. Second, the simulation considers only a single asset and does not account for things like how other stocks are moving or even macroeconomic news events. Third, although the exponential offset distribution produces realistic depth profiles, it remains a stylized version of actual order-placement behavior shown by the traders. Finally, the current implementation uses discrete time steps rather than continuous-time Poisson arrivals, which may affect the precise timing statistics of order flow.

Conclusion

In this study a fully stochastic, reproducible limit order book simulator is created to analyze the behaviour of bid-ask spread under different market conditions. After running five main experiments and three parameter sweeps the data made it very clear that liquidity is the biggest factor in how wide the spread gets. It was observed whenever the liquidity was low the spread was not just wider but unpredictable and volatile. Order-flow imbalance and high cancellation rates act as amplifiers making the spread fluctuations even more intense, while the simulator successfully captured the real world patterns like the rightly skewed spread data and the negative link between liquidity and the costs. This confirms that even a simple model can replicate the complexities of real financial markets.

The results provide a solid computational backing for the theoretical predictions in market microstructure theory by Amihud et al. (2005); Chordia et al. (2002) and align with what we see in actual high-frequency data as documented by Bouchaud et al. (2002); Cont et al.

(2010). By combining a dynamically changing mid price with realistic order placements and cancellation rules, the simulator offers a transparent and controllable lab for testing specific causes of market behaviour that are very difficult to isolate in real-market data.

The most important contribution of this study is how transparent and easy it is to replicate real financial markets. All code, parameters, raw data files, and analysis scripts are available upon request, which enables other researchers to replicate the work or even use the data model for their own data calibration. This open science approach is really needed in the field of computational finance to help close the gap between simple theories and messy realities of the market.

In summary, the simulator proves that there is no need for a massive complex system to analyze realistic market dynamics; instead a zero intelligence model also works if rules are similar to real world behaviour. This framework is highly flexible and not only serves us but also becomes a starting point for studying things like high frequency trading or even the complexities of how an order moves in a market.

Table 1

Summary of simulation results

Experiment	Mean Spread	Std. Dev.	Median Spread	Min Spread	Max Spread	Mean Mid-Price	Total Trades	Corr(Spread, Liquidity)	Corr(Spread, Imbalance)
Balanced Market	0.1914	0.1339	0.17	0.00	0.82	99.9973	64,134	-0.3268	0.2514
Buyer-Heavy (70%)	0.1329	0.2089	0.09	0.00	9.02	104.3665	61,186	-0.2972	-0.3339
Seller-Heavy (70%)	0.1411	0.1675	0.11	0.00	5.87	95.8925	61,204	-0.3279	-0.3782
High Cancellation	0.2362	0.2029	0.18	0.00	2.02	99.2906	63,132	-0.4014	-0.2529
Low Liquidity	0.4769	0.4719	0.35	0.00	8.14	102.3437	24,560	-0.1294	0.0961

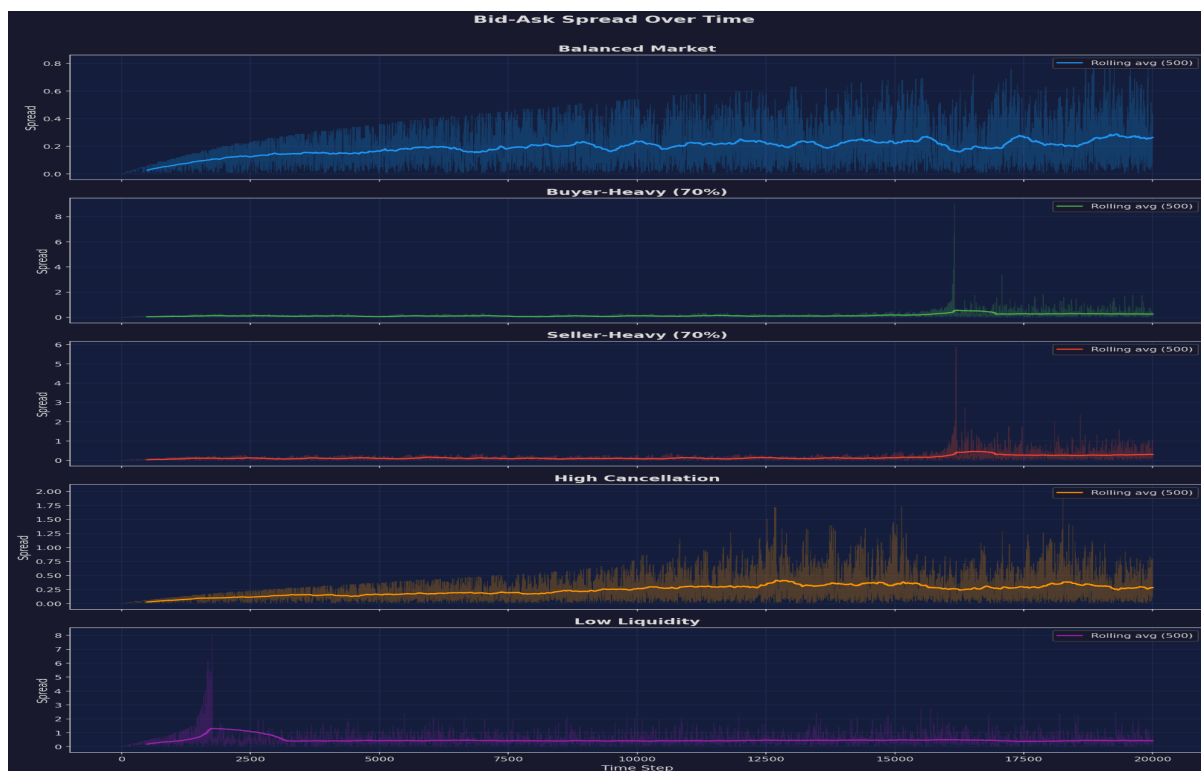


Figure 1

Bid-ask spread evolution over time

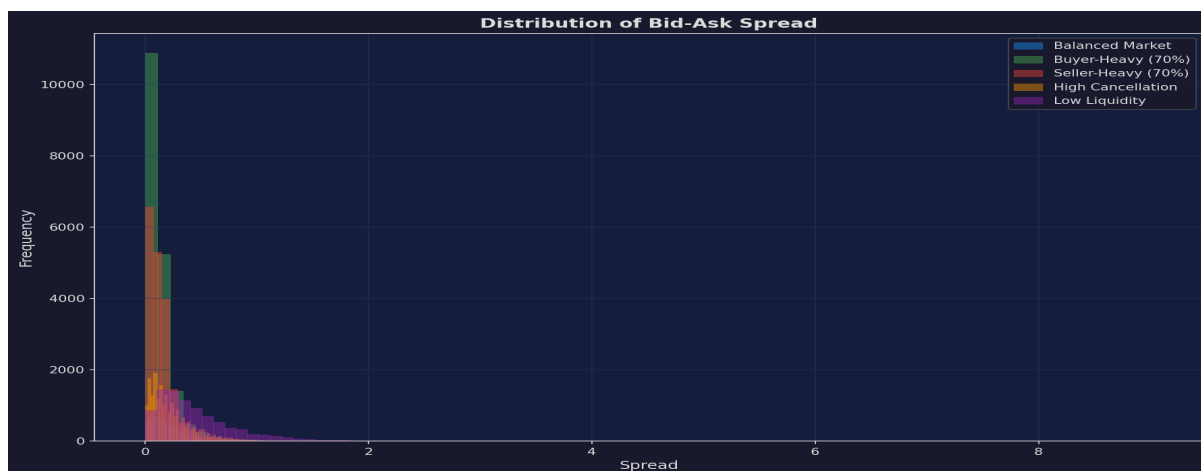


Figure 2

Distribution of Bid-ask spread across experiments

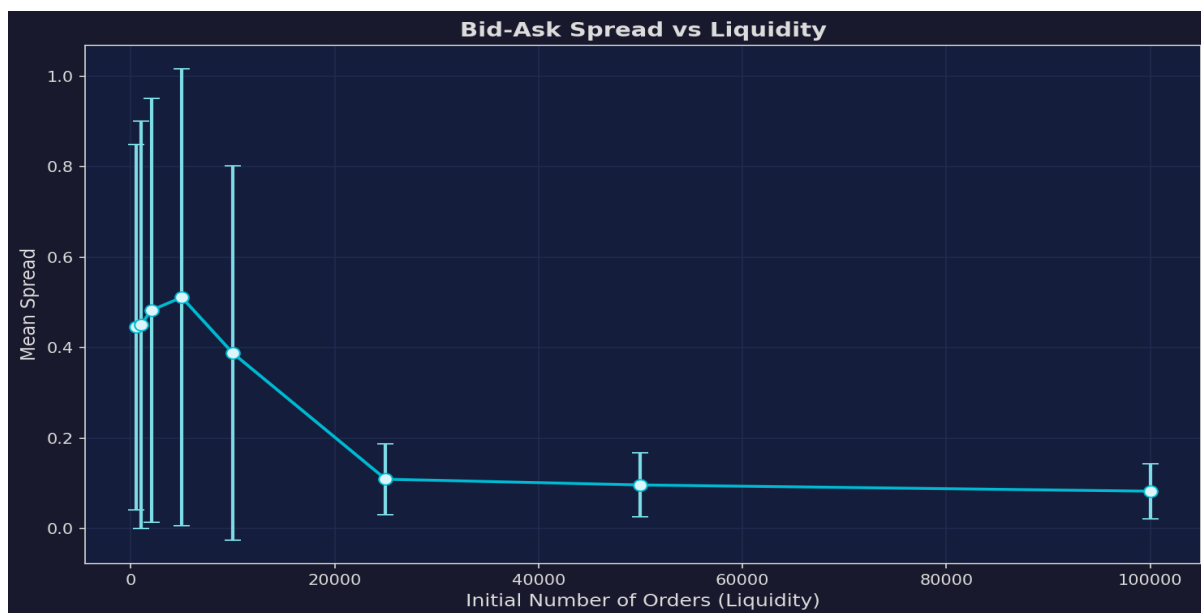


Figure 3

Mean bid-ask spread as a function of initial liquidity

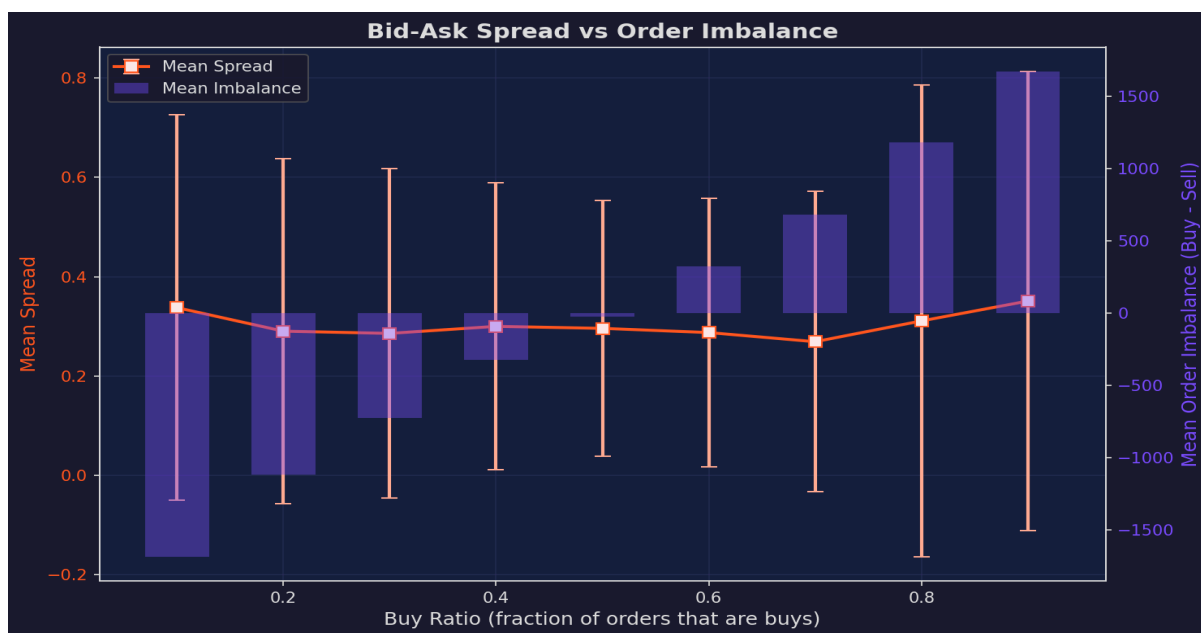


Figure 4

Mean bid-ask spread as a function of order-flow imbalance

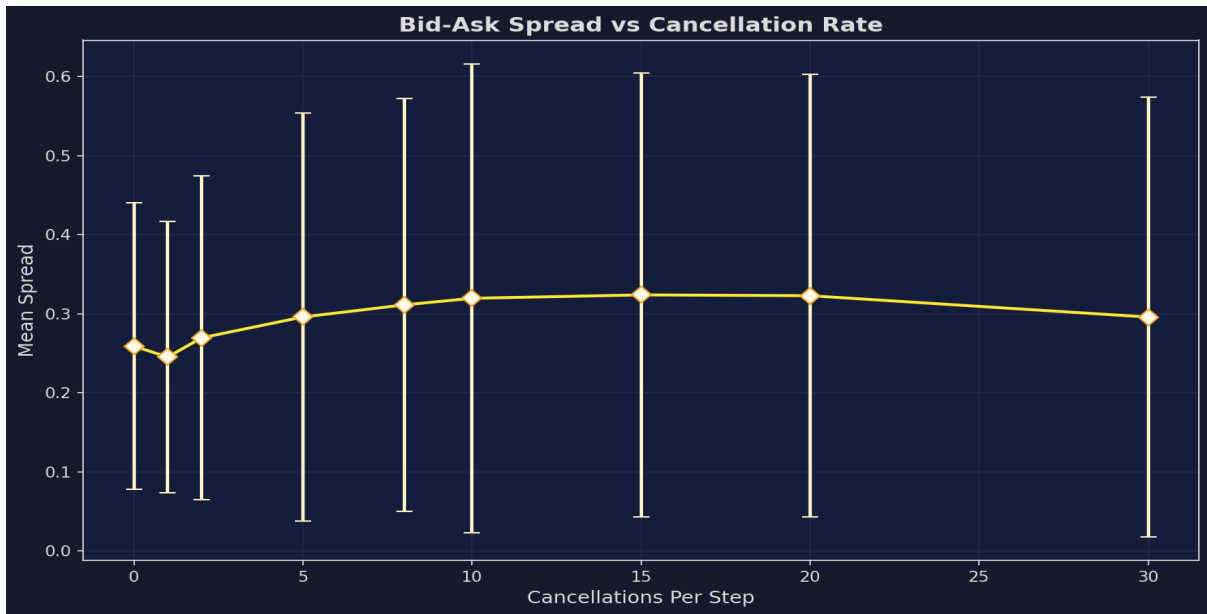


Figure 5

Mean bid-ask spread as a function of cancellation rate



Figure 6

Mid price evolution over time across experiments

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